

Senior Research Project

Charter School of Wilmington Class of 2027

Introduction to Senior Research:

Welcome to your final year at the Charter School of Wilmington! The Senior Research Project is a graduation requirement designed to prepare you for the type of independent thinking, research, and communication you will be expected to complete in college and beyond.

For this project, you will explore a topic you are genuinely passionate about. Your topic may fall within science, engineering, medicine, technology, public health, environmental studies, ethics, policy, or another approved academic field. The most successful projects are driven by curiosity, personal interest, and real-world relevance.

You may complete this project:

- Individually
- With one partner (pair)
- In a group of up to three students

***Note:** Regardless of group size, all students are expected to contribute meaningfully to the research, writing, and presentation.*

Through this year-long project, you will:

- Ask thoughtful, real-world questions
- Investigate a topic that matters to you
- Apply analytical and critical-thinking skills
- Collect and interpret evidence or data
- Communicate your work through a formal paper and a professional presentation

This document outlines the expectations, policies, and key deadlines for Senior Research. Please review it carefully with your parent/guardian. A signed acknowledgement form & an initial proposal must be submitted to **Dr. Valenti** by **2:10 PM on Monday, April 20**.

***Note:** All checkpoints, rubrics, and submission links are posted in the Senior Research Schoology course. Schoology should be your primary reference throughout the year.*

Collaboration & Group Expectations:

Students may work in groups that range from **one person to three people**. However, once the group size is chosen and the research proposal is submitted and confirmed, it cannot be adjusted later. Students should take care to choose partner(s) that align with their research goals and with whom they can work throughout the school year with minimal conflict.

Group work is encouraged for projects that benefit from collaboration; however:

- Each student is still responsible for active participation
- Individual contributions may be assessed separately
- All group members must be able to explain all aspects of the project during the presentation and Q&A.

***Note:** Choosing to work in a group does **not** reduce expectations—it increases the importance of communication, planning, and accountability.*

Research Pathways:

Your research topic should align with one of the following pathways. These options are intentionally broad to allow flexibility and creativity.

1. **Empirical Research:** Investigate a question through data collection, experimentation, or observational study.
2. **Invention/Engineering Design:** Design, build, and test a solution or prototype that addresses a real-world problem.
3. **Research Assistantship:** Conduct research in collaboration with a university, laboratory, organization, or professional mentor.
4. **AP Seminar AND AP Research:** A two-year program focused on inquiry, academic writing, and formal presentation around a topic of personal interest.

Your research advisor will help ensure that your topic is appropriate in scope, academically rigorous, and feasible.

Topic Selection Guidance:

Your topic should:

- Be something you are genuinely interested in or curious about
- Allow for deep investigation, not just a summary
- Connect to real-world issues, applications, or unanswered questions
- Be appropriate for high-school-level research with college-level expectations.

If you are unsure whether a topic is appropriate, ask early. Topic approval is required before moving forward, and expectations will be high when it comes to the level of investigation and real-world application. It is highly recommended that students think about their future college goals and potential majors as they choose their topic.

In approving your research proposal, advisors will consider the quality of your investigation, the applicability, and the very high expectations CSW has for seniors. Therefore, any changes made to your study after approval must be sent to and approved by the Research Coordinator. Changes that reduce the caliber of research will need to be justified to the Research Coordinator and your appointed advisor.

Research Expectations & Resources:

You are expected to use credible, peer-reviewed sources. Approved resources include:

- **UDLib/SEARCH** – Statewide academic databases including JSTOR, EBSCOhost, and ScienceDirect.
- **Google Scholar** (<https://scholar.google.com>) – Peer-reviewed articles and academic books.
- **PubMed** (<https://pubmed.ncbi.nlm.nih.gov>) – For biomedical research topics.
- **IEEE Xplore** (<https://ieeexplore.ieee.org>) – For engineering and technology topics.
- **National Center for Biotechnology Information (NCBI)** – (<https://www.ncbi.nlm.nih.gov/>)
- **Delaware Public Library Databases** – Accessible with a public library card.

***Do NOT use general websites or Wikipedia as primary sources. However, Wikipedia may help you find legitimate primary sources via its reference lists. All APA formatting should follow the 7th edition guidelines. Use Purdue OWL (<https://owl.purdue.edu>) for formatting help.*

AI Tools Usage Policy:

- AI-based writing assistants (like ChatGPT, Grammarly) may be used for editing and feedback only—NOT for generating full content.
- You must disclose in your final submission if AI tools were used for editing.
- Generated AI content submitted as original writing will be treated as plagiarism.

APA Format and Research Paper Components:

1. **Title Page**
2. **Abstract**
3. **Introduction** (background research, purpose, and hypothesis)
4. **Methods** (materials, variables, controls, sample size)
5. **Results** (graphs, tables, statistical analysis)
6. **Discussion** (interpretation, limitations, future directions)
7. **References** (APA style)
8. **Appendices** (if applicable)

Checkpoint Calendar (2026-2027):

Checkpoint	Due Date	Description
Checkpoint #1: Initial Proposal	April 20, 2026 (end of school day)	Signed acknowledgment, 2-page introduction, hypothesis, methods, references
Advisor Appointed	End of Junior Year	A research advisor will be matched to you based on your topic and its alignment with their expertise or professional background.
Checkpoint #2: Experimental Results	December 1, 2026	Notebook, data, sample size justification, statistical methods
Checkpoint #3: Draft Paper	January 24, 2027	Complete draft in APA format
Checkpoint #4: Final Paper & Turnitin Submission	March 21, 2027	Final paper with Turnitin.com originality report
Checkpoint #5: Senior Research Presentations	End of 3rd Marking Period (Senior Year)	Formal presentations with poster

Data and Statistical Expectations:

- Minimum 30 data points per group (control & variable).
- Use of tools such as Google Sheets or Excel for analysis.
- Apply appropriate statistical tests (e.g., t-test, ANOVA, regression)
- All results must include clear graphs or tables
- Consult math teachers or research advisors for statistical support.

Safety & Human Subject Research:

- Follow updated CSW Safety Manual guidelines, including PPE and lab protocols.
- Human subject research requires informed consent forms.
- Projects involving vertebrate animals, biohazardous materials, or sensitive materials require pre-approval from the **Research Coordinator**.

Plagiarism & Turnitin Policy:

- All papers must be submitted to Turnitin.com.
- Final submission must include the originality report.
- Plagiarism—including AI-generated writing—will result in project failure and disciplinary action.
- Citation support: <https://www.citationmachine.net/apa>

Presentation Guidelines:

- Poster.
- Business casual or school-appropriate professional attire required
- Be prepared to submit:
 - Final Paper
 - Poster
- A formal Q&A session will follow your presentation.

Getting Help & Support:

- Senior Research Schoology Course
- Library databases via UDLib/SEARCH
- Research Advisor Office Hours
- [Purdue OWL](#) (APA formatting)

Why Senior Research Matters?

Senior Research is more than a graduation requirement—it is one of the most meaningful academic experiences you will complete at CSW. This project demonstrates to colleges that you can manage a long-term commitment, think independently, and engage deeply with a topic you care about. It appears on your transcript as its own category, signaling a sustained research experience, and often becomes a strong talking point in college essays, interviews, and recommendation letters. Most importantly, Senior Research helps you build skills—critical thinking, research, writing, data analysis, and presentation—that you will use in college and beyond, while giving you the opportunity to explore a topic that genuinely matters to you.

Acknowledgment of Senior Research Form

Acknowledgment Form by **2:10 PM on Monday, April 20, 2026**

We have reviewed the Senior Research requirements and due dates together.

Student's name (please print): _____

Student's signature: _____

Parent's signature: _____

Date: _____